



HK IMO Training 2021-2022
Problem Set 5
For 23 October 2021



Problem 17. Let p be an odd prime. Initially, we have two integers chosen from $S = \{1, 2, \dots, p-1\}$. In each step, if the current integers we have are a and b , we can replace a by the unique integer $c \in S$ such that $bc \equiv a \pmod{p}$, or replace b by the unique integer d such that $ad \equiv b \pmod{p}$. Prove that we can make the two integers the same after finitely many steps.

Problem 18. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)^2 + f(y^2)) = (x - y)f(x - f(y)).$$

Problem 19. Let $ABCD$ be a trapezoid inscribed in a circle Γ , where $AD \parallel BC$. The rays AB and DC intersect at E . Let F be the point on Γ such that BF bisects AD . The circumcircle of $\triangle BEF$ intersects the ray BC again at P . Prove that $\angle EDP = 90^\circ$.

Problem 20. Let n and k be positive integers. Two players X and Y play a game on a $2n \times 2n$ chessboard as follows. Firstly, X chooses one of the cells secretly without letting Y know which cell it is. We call this the position of X . In each round, the following things happen in order.

- X moves to an adjacent cell secretly. Note that an adjacent cell to C means a cell sharing a common edge with C which is different from C .
- Y chooses k cells. If one of the cells chosen by Y is the position of X , then Y wins the game.

Find the smallest possible value of k such that Y can always win the game after a finite number of rounds.